

Thomas VAUDESCAL

Quantitative Developer | Quantitative Researcher

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Profile

Quantitative developer with an MSc in Financial Engineering from HEC Montréal (**GPA 3.88/4.3**). Builds backtesting, pricing and trading systems in **Python, C++ and Rust**, and takes strategies from research to live trading. Founder of Synerqo, previously at National Bank Financial Markets and Deloitte.

Professional Experience

Synerqo

Montreal, Canada

Founder & Quantitative Developer

May 2025 - Present

- Built an **event-driven options backtesting engine in Python and Numba** that models spreads, slippage, commissions, margin and assignment. It runs five years of SPX intraday option chains (**216M rows**) in **under a minute** on one workstation, with 2,448 automated tests.
- Designed, backtested and **deployed a systematic SPX options strategy to live trading**, turning MSc research on dealer gamma exposure into a production signal and execution workflow.
- Built an **options and structured products pricing engine**: Monte Carlo, PDE solvers, Heston, SABR and SVI calibration, and Greeks via **adjoint algorithmic differentiation (AAD)**, 2.9x faster than bump-and-reprice.
- Developed **Rust execution components** for market-data ingestion, quote updates and order routing, and profiled their latency against the Python research stack.
- Delivered client software: **Ganesha**, a retrieval-augmented (RAG) AI assistant in production, and a custom CRM for an insurance client that **saves it tens of thousands of dollars**.
- **Technologies**: Python, Numba, JAX, C++, Rust, FastAPI, PostgreSQL, TypeScript, QuantLib, QuantConnect.

Deloitte

Montreal, Canada

Quantitative Analyst | Financial Risk

October 2024 - June 2025

- Developed and validated **credit risk models (PD, LGD, EAD)** for major Canadian financial institutions under IFRS 9 and Basel III/IV.
- Tested model assumptions and calibration through statistical backtesting, sensitivity analysis and stress testing.
- Built **automated validation pipelines and monitoring dashboards in Python**, cutting manual review time by **40%**.
- Presented findings to **senior stakeholders and client risk committees** (OSFI context).
- **Technologies**: Python, R, SQL, statistical modeling.

National Bank Financial Markets

Montreal, Canada

Quantitative Developer Intern | Global Equity Derivatives R&D

May 2023 - December 2023

- Built an **end-to-end Python backtesting framework**: signal generation, portfolio construction, transaction costs, risk analytics and performance attribution.
- Developed and deployed low-volatility and momentum **SmartBeta strategies covering \$800M in AUM** within a \$7B structured products division, with constrained portfolio optimization in cvxpy.
- Delivered **\$1M in annual cost savings** by moving strategies from an external vendor to internal production systems.
- Worked with **structuring and trading teams** to integrate quantitative signals into pricing and hedging workflows.
- **Technologies**: Python, cvxpy, SQL, Bloomberg Terminal & API, AWS (EC2, S3).

Videns Analytics

Montreal, Canada

Data Analyst

April 2020 - June 2021

- Led **insurance analytics consulting engagements**: data cleaning, predictive modeling and automated reporting.
- Designed a **Python, statistics and machine learning curriculum** adopted by UQAC for a university course. Promoted from intern to full-time.

Technical Skills

Programming	Python, C++, Rust, SQL, R, MATLAB, TypeScript.
Computer Science	Object-oriented design, design patterns, data structures and algorithms, event-driven architecture, concurrency (asyncio), performance optimization, latency profiling.
Python Stack	NumPy, Numba, JAX, pandas, Polars, SciPy, scikit-learn, PyTorch, cvxpy, QuantLib.
Quant Finance	Backtesting, walk-forward validation, derivatives pricing, Greeks via AAD, Monte Carlo, PDE methods, volatility calibration (Heston, SABR, SVI), portfolio optimization, market microstructure.
Research Methods	Pre-registered hypotheses, deflated Sharpe ratio, probability of backtest overfitting, bootstrap, statistical testing.
Risk Modeling	Credit risk (PD, LGD, EAD), model validation, stress testing, model monitoring.
Tools	Git, Linux, Docker, AWS (EC2, S3), DuckDB, PostgreSQL, FastAPI, REST and WebSocket APIs, Bloomberg Terminal & API.
AI & Web	RAG, LLM integration, Qdrant, React, Next.js.

Education

HEC Montreal

Montreal, Canada

MSc in Financial Engineering (Thesis) | GPA 3.88/4.3

August 2022 - December 2024

- **Thesis:** Gamma Exposure and Intraday Volatility Dynamics in the SPX Options Market. Graded **A+ with commendation from the examination committee**, now a working paper for the International Review of Financial Analysis. Applied to a live systematic options strategy.
- **Scholarships of excellence:** Deloitte M.Sc., Alpha Fixe Capital, Jocelyne & Jean C. Monty I and II.
- **Distinctions:** National Bank challenge winning team at ConUHacks VIII (2024), top 10 of 66 teams at the McGill FIAM Portfolio Management Hackathon, 2nd place at the HEC Montreal Trading Competition.
- **Leadership:** Vice President Research at the Trading Club and R&D Analyst at the BNI-HEC Investment Fund (\$5M AUM).
- **Selected coursework:** Stochastic Calculus, Derivatives Pricing, Financial Econometrics, Numerical Methods, Machine Learning for Finance, Risk Management.

HEC Montreal

Montreal, Canada

BBA | Mathematics, Finance and Economics

August 2017 - May 2022

- **GPA:** 3.8/4.3 in specialization courses. Quantitative track combining mathematics, economics and finance.
- **Leadership & teaching:** Vice President Education at the Data Science Committee (Python workshops, machine learning seminars), Teaching Assistant for Intermediate Microeconomics (200+ students).

University of Montreal

Montreal, Canada

Minor in Mathematics

August 2018 - August 2022

- Probability Theory, Mathematical Statistics, Stochastic Processes, Linear Algebra, Real Analysis, Programming (C/C++).

Languages

French	Native
English	Professional proficiency (spoken and written)

Interests

Technology	Personal quant projects on GitHub: quant-paper-lab (finance papers reproduced on public data), fx_strategies (walk-forward framework), options market-data pipelines.
Intellectual	Chess (2000 Elo) , mathematics, philosophy.
Sports	Track and field, marathon running, mountain hiking, indoor climbing, swimming, weight training, scuba diving.